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Galactic panel

Matthias R. Schreiber | matthias.schreiber@usm.cl

USM

faculty

Towards a volume-limited sample of cataclysmic binaries

Abstract

Close white dwarf binaries produce supernova Ia explosions and represent the dominant gravitational wave sources to be detected with future space interferometers. Despite their importance, we still struggle to understand their formation and evolution and are therefore unable to make reliable predictions of their relative numbers and detailed characteristics. Our ignorance of the evolution of white dwarf binaries is related to the absence of large, unbiased and well-characterized samples to calibrate evolutionary theories. This proposal is part of a large scale project aiming at establishing a representative volume limited sample of cataclysmic variables (CVs), the most abundant type of white dwarf binaries. This is now possible thanks to the eROSITA survey which is able to detect the X-ray emission of all CVs up to 500 pc. The role of the Speculoos observations proposed here is to measure the orbital period of CVs discovered by eROSITA through photometric variability.

Observing Blocks

Instrument/Telescope	Req. time	Min. time	1 st Option	2 nd Option
CCD Speculoos/Speculoos	19 nights	10 nights	Any Any	Any Any

Cols

Name	Institution	e-mail	Observer?
Jaco Brink	OnCL	jbrink@aip.de	False
Axel Schwope	OnCL	aschwope@aip.de	False
Sarai Zapata	USM	sarai.zapata@usm.cl	False

Status of the project

- Past nights: 0
- Future nights: 141
- Long term: True

- Large program: False
- Thesis: False

List of Targets

ID	RA	DEC	Mag
Em01004159	00:07:43	-69:59:48	19.2
Em01011138	00:47:08	-48:45:22	18.6
Em01027135	01:50:12	-45:21:20	18.2
Em01028138	01:51:52	-47:38:27	18.7
Em01035141	02:13:45	-51:16:55	19.7
Em01048117	03:16:22	-27:59:10	19.5
Em01060135	03:56:47	-44:48:20	16.7
Em01066114	04:26:34	-25:02:28	15.0
Em01067126	04:24:06	-34:41:17	19.3
Em01071168	04:49:38	-77:56:03	18.2
Em01077099	05:07:41	-09:13:38	18.6
Em01088075	05:47:27	+15:40:51	19.2
Em01092117	06:05:36	-25:59:47	19.9
Em01100126	06:45:02	-37:07:51	19.0
Em01119138	07:50:24	-48:10:35	17.4
Em01121156	08:01:39	-64:44:01	18.7
Em01125114	08:17:50	-23:22:02	18.9
Em01143123	09:29:41	-34:11:13	20.4
Em01164141	10:46:13	-51:18:12	17.8
Em01173141	11:39:28	-52:23:20	16.6
Em01188132	12:31:12	-42:35:20	16.8
Em01190156	12:36:33	-66:46:02	17.8
Em01191093	12:38:14	-03:39:34	17.7
Em01191150	12:41:29	-60:58:20	18.6

SCIENTIFIC AIM AND RATIONALE

Cataclysmic variables (CVs) are the most common endproducts of close binary evolution (1). They consist of a white dwarf (WD) primary that accretes matter via Roche-lobe overflow (RLOF) from a low-mass companion (9). More than 90% of the systems have binary orbital periods below 4 hours and, in very general terms, they evolve under the influence of angular momentum loss (AML) through magnetic braking and gravitational radiation, from long to short orbital periods. Understanding close binary evolution is of imminent relevance to uncover pathways to SNIa and to understand the foreground signal of the planned space-based gravitational wave observatory. Yet there are fundamental unsolved problems concerning their sample properties and their evolution that shall be tackled by this proposal.

Accretion onto the WD is strongly dictated by its magnetic field. In CVs where the WD has a weak magnetic field an accretion disk forms that may extend to the surface of the WD, while in CVs with highly magnetized WDs (magnetic CVs, mCVs), known as polars and intermediate polars, there is at most a truncated disk and matter flows via accretion columns onto the magnetic poles of the WD. Observationally mCVs can be distinguished from non-magnetic systems by their emission lines profiles (symmetric disk lines for non-magnetic CVs, asymmetric lines from streams in mCVs), the strength of the He II $\lambda 4686$ emission line (similar to that of $H\beta$ in mCVs, weak or absent in non-magnetic CVs), and the occasional detection of cyclotron harmonic emission lines or Zeeman absorption lines as direct indicators of magnetism. Photometrically, most CVs show periodic variability on the binary orbital period (unless the orbital inclination is too small, which applies to a very small subset of the known population) and may show periodic variability also on the spin period of the accreting WD ($P_{\text{orb}} = P_{\text{spin}}$ for the polars, $P_{\text{spin}} \sim 0.1P_{\text{orb}}$ for the intermediate polars, flickering variability for the non-magnetic CVs.)

The evolutionary status of a given single system is written into its binary orbital period and the evolution of the class (and the sub-classes) is written into the binary orbital period distribution. Thus, the binary orbital period is *the* fundamental parameter to be determined for individual systems and complete samples to fully understand binary evolution.

There are two fundamental issues that were hampering progress in the past, one on the theoretical, one on the observational side. In the past, binary population synthesis (BPS) models (e.g. by 3) have completely neglected to take into account the effect of magnetic fields. It was shown by Belloni et al. (2), that magnetic fields will play an integral role in the evolution of compact binaries by weakening the effect of magnetic braking, and thereby slowing down the angular momentum loss (AML) in mCVs. Observationally only recently it has become clear that mCVs and non-magnetic systems have significantly different orbital period distributions (6), which implies that the strength of the magnetic field plays a significant role and forces mCVs to have different evolutionary pathways from non-magnetic systems. However, the strength of AML is not fully quantified so that a full-scale BPS that could be compared with observations is still missing.

On the observational side, sufficiently large complete samples are not yet available so that neither the space density nor the fraction of mCVs among all CVs are known. The space density has uncertainties of a factor of 2 or larger which is simply due to the small sizes of either flux- or volume-limited samples (7; 5). Pala's sample consists of 42 CVs, has an estimated completeness $>80\%$, and comprises a fraction of $\sim 36\%$ mCVs. However, a first eROSITA-selected flux-limited sample with 23 CVs contained only 2 magnetic systems (8).

The prospects to overcome the current observational limitations are excellent. All CVs are X-ray emitters (4) and this property allows to build large and complete volume-limited samples from the exploitation of the sensitive X-ray all-sky survey performed by eROSITA. From a probabilistic source match of eROSITA X-ray sources with Gaia objects a first batch of high-confidence CVs was selected. The target properties that led to their selection are described and illustrated in Fig. 2.

IMMEDIATE OBJECTIVES

From the analysis of existing EFOSC2@NTT spectroscopy of 150 candidates the target association could be quantified to be better than $> 90\%$, as 133 new CVs were confirmed. eROSITA data allows to build a volume-limited sample of the various CV sub-classes out to 500 pc. The whole sample will comprise of about 1000 systems. From the analysis of the confirmed new CVs two subsets were selected, one within 500 pc, and another magnetic one without volume-limitation. These are labeled with priorities 1 and 2 in the target list below. The first sub-set is chosen to build the desired volume-limited sample, the second subset to sample the orbital period distribution of mCVs with better statistics than in (6). All CVs are new to the literature, they lack knowledge about their orbital periods. These can most easily be obtained from dedicated time-resolved photometric observations, that is proposed here.

We propose to perform time-resolved photometric observations of new confirmed CVs for which nothing more than their X-ray properties and an identification spectrum are known. An example is shown in Fig. 1. The strong HeII emission line at $4686\text{\AA}$ and the pronounced broad cyclotron harmonic (FWHM $\sim 300\text{--}400\text{\AA}$) centered on $5400\text{\AA}$ clearly identify the object as a strongly magnetic CV. From the analysis of existing ULTRACAM@NTT photometry, lasting about 100 min, an eclipse of the accreting mWD by the low-mass donor (not detected in the identification spectrum) was found. The observation was too short to uncover the orbital period. The determination of the binary orbital periods of all the objects listed below is the one and only aim of this proposal. This measurement will deliver the missing piece of information to make use of the newly discovered CVs for sample studies and evolutionary considerations.

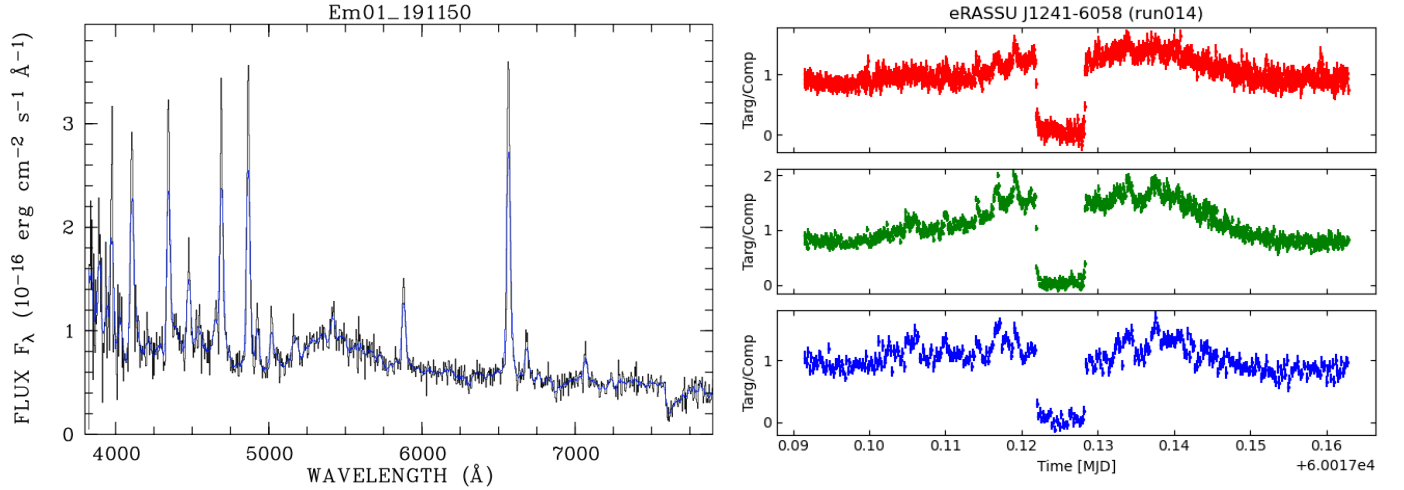


Figure 1: NTT identification spectrum of one of the proposed targets (left). The object is uniquely identified as mCV through its strong HeII line and the cyclotron hump at $5400\text{\AA}$. High-time resolution photometry with ULTRACAM@NTT of the same target reveals an eclipse of the WD (right)

References

- [1] Belloni D., Schreiber M. R., 2023, in , Handbook of X-ray and Gamma-ray Astrophysics. Edited by Cosimo Bambi and Andrea Santangelo. p. 129, [doi:10.1007/978-981-16-4544-0_98-1](https://doi.org/10.1007/978-981-16-4544-0_98-1)
- [2] Belloni D., Schreiber M. R., Pala A. F., Gänsicke B. T., Zorotovic M., Rodrigues C. V., 2020, *MNRAS*, **491**, 5717
- [3] Goliasch J., Nelson L., 2015, *ApJ*, **809**, 80
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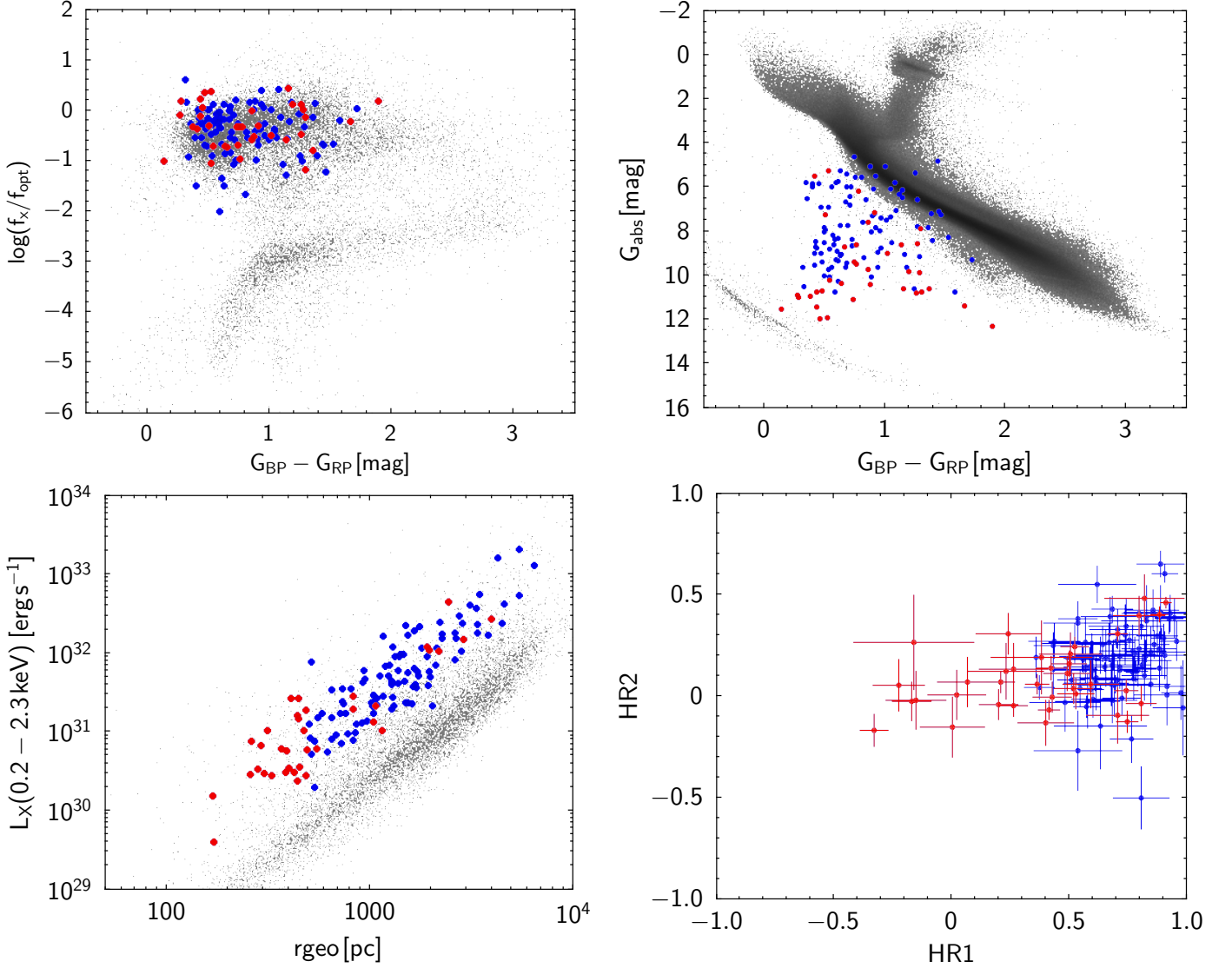


Figure 2: Diagnostic diagrams of newly identified southern CVs from eROSITA. Shown are an X-ray/optical color-color diagram, a color-magnitude diagram, a luminosity-distance diagram, and an X-ray color-color diagram. Shown in blue are all new CVs, those selected for this proposal are shown in red. The X-ray/optical color-color diagram shows the comparably high X-ray flux of the CVs (as high as those of AGN and 2-3 orders of magnitude higher than stars, both shown as grey background sources). The CMD illustrates the large spread in absolute magnitude thus covering all kinds of CVs (the grey background is made of non-X-ray emitting unrelated objects to outline the ZAMS and the WD sequence). The Lx-dist diagram illustrates the potential to build large complete samples out to 500 pc (grey background objects are selected from deeper eROSITA surveys under investigation in e.g. SDSS), and the X-ray hardness ratio diagram separates mCVs from non-magnetic ones through their soft X-ray spectra with $\text{HR1} < 0.3$. A hardness ratio is defined as $\text{HR} = (H - S) / (H + S)$, with H and S being X-ray photons in two adjacent energy bands. With the given normalization, HR-values are constrained within +1 and -1. An HR1-value above 0.3 is compatible with just thermal emission from an accretion column or the disk-WD boundary layer. A value below 0.3 indicates the presence of a soft, blackbody-like source, as discovered from the heated photospheres in the vicinity of the accretion regions in mCVs.

TECHNICAL DESCRIPTION

For each of the targets in the list below we request time-resolved photometry during two successive nights with SPIRIT@SPECULOOS lasting 4 hours each. We choose the r filter for its high sensitivity. We plan to use integration times of 60 sec (for objects brighter than $G=17.5$ we will use 30 sec exposures). The observations shall be performed without interruption for the planned four hours per night. The first observation will give a first estimate of the binary orbital period. A second observation in the following night is requested to then determine the period with about 10% accuracy, sufficient for the later analysis in an evolutionary context. The second observation has to be placed in the following night after the first to not loose the cycle counts.

The target list is compiled for observations beginning in October 2024, the observations shall last 6 month. The list comprises 24 objects. For each object we request 8 hours which adds up to 192 hours or 19 nights. While this would significantly slow down our progress, we could start the project with a smaller number of targets and slightly reduced observing time. However, allocating less than 10 nights (corresponding to ~ 13 targets) would not make sense in the light of our ambitious goals. **Given the large range of right ascensions covered by our target sample, the requested 19 nights should be more or less homogeneously spread over the observing period (October-March).**

Name	RA (deg)	DEc (deg)	G (mag)	distance	priority
em01_004159	1.93038	-69.99656	19.2	488	1
em01_011138	11.78195	-48.75615	18.6	1963	2
em01_027135	27.55063	-45.3555	18.2	282	1
em01_028138	27.96612	-47.64097	18.7	367	1
em01_035141	33.43588	-51.28197	19.7	451	1
em01_048117	49.08988	-27.98619	19.5	832	2
em01_060135	59.19388	-44.80556	16.7	170	1
em01_066114	66.64025	-25.04119	15.0	263	1
em01_067126	66.02462	-34.68814	19.3	472	1
em01_071168	72.40904	-77.93408	18.2	831	2
em01_077099	76.91989	-9.22715	18.6	483	1
em01_088075	86.86132	15.68083	19.2	290	1
em01_092117	91.40183	-25.99629	19.9	2883	2
em01_100126	101.25763	-37.13094	19.0	447	1
em01_119138	117.59877	-48.17633	17.4	2432	2
em01_121156	120.41276	-64.73352	18.7	1065	2
em01_125114	124.45917	-23.36711	18.9	2183	2
em01_143123	142.42000	-34.18694	20.4	425	1
em01_164141	161.55500	-51.30347	17.8	491	1
em01_173141	174.86792	-52.38894	16.6	1902	2
em01_188132	187.80042	-42.58897	16.8	439	1
em01_190156	189.13583	-66.76711	17.8	407	1
em01_191093	189.55667	-3.65936	17.7	171	1
em01_191150	190.37167	-60.97225	18.6	442	1

JUSTIFICATION OF LONG-TERM STATUS OR WORK PLAN FOR LARGE PROGRAMS.

The current application is based on the analysis of a first batch of CVs from existing EFOSC2@NTT observations. These were selected on the basis of only the first of four complete X-ray all-sky surveys with eROSITA (Merloni et al, 2024, A&A 682, A34). The deeper data were stacked, counterparts were associated and a large-scale spectroscopic identification program was implemented in SDSS-V (Almeida et al., 2023, ApJS 267, 44). These SDSS observations are in full swing. Contrary to earlier phases of the SDSS survey, SDSS has gone all sky though modifications of the Las Campanas 2.5 m DuPont telescope. This is particularly relevant in the context of this proposal as it delivers southern targets. The southern CV sky is much less explored than the northern sky and many new CVs are to be discovered. All the new discoveries will have eROSITA X-ray and spectroscopic information but they are lacking detailed photometry that would allow the determination of their orbital periods. The Milky Way Mapper program of SDSS-V, that will run until 2027, is supposed to observe 1087 southern CV candidates brighter than $G = 20$ and within 500 pc. We estimate the completeness fraction to be 40% max. About half of the objects will be known or will have orbital period information from elsewhere, hence we will select of order 200 CVs that need time resolved photometry in the same manner as described here. This adds up to an equivalent of 160 nights to be distributed over the next 4 years. The result will be a cleanly characterized complete sample to constrain the evolutionary channels of accreting compact white dwarf binaries.

CURRENT STATUS OF THE PROJECT

In the last few years research performed by our group on magnetic white dwarfs and white dwarf binary evolution had a global impact. We showed that the evolution of magnetic white dwarfs in cataclysmic variables can be understood by incorporating reduced magnetic braking in the standard prescriptions for the evolution of these systems (1). We furthermore demonstrated that one of the scenarios frequently suggested for the generation of magnetic fields in white dwarfs, the common envelope dynamo model, fails to explain the observations of detached and semi-detached white dwarf binaries (2). We also published a comprehensive review on the X-ray emission of magnetic cataclysmic variables in ApJS (3) and a new idea for the magnetic field generation in white dwarfs in close binary stars in Nature Astronomy (4). Our partners are running a highly successful identification program of compact sources from the eROSITA surveys (5) which led to the discovery of the 2nd white dwarf pulsar (6) and, most importantly, allows to establish a large and volume limited sample of accreting white dwarf binaries. We have started a large-scale follow-up programs of these systems with the SDSS (7) and ESO telescopes. We here propose to measure the orbital period of systems discovered by eROSITA and confirmed through spectroscopy. The orbital period distribution of these systems is key to understand their evolution (8). Throughout the next four years we will combine eROSITA and SDSS discoveries of new CVs (led by Schwöpe and Schreiber) with follow-up observations (performed and analyzed by two PhD students, Jaco Brink from AIP and Sarai Zapata from UTFSM). This project plays a crucial role in the development of white dwarf binary research in the region of Valparaíso. In particular, combining detections of the European X-ray satellite (eROSITA) with follow-up observations using telescopes located in our country represents a fantastic win-win situation.

Relevant publications

- [P1] D. Belloni, M. R. Schreiber, A. F. Pala, B. T. Gänsicke, M. Zorotovic, and C. V. Rodrigues, “Evidence for reduced magnetic braking in polars from binary population models,” *MNRAS*, vol. 491, pp. 5717–5731, Feb. 2020.
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